Alejandro Rodriguez

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SKILLS

Programming: Python, MATLAB, pandas, scikit-learn, Matplotlib, NumPy, Seaborn, feature engineering, statistical modeling. LaTeX
Modeling: Mathematical modeling, computational simulation, numerical methods, optimization, stochastic processes, Markov chains
Engineering: MATLAB Antenna Toolbox, antenna simulation, beam pattern analysis, S-parameter analysis, orbital mechanics

WORK EXPERIENCE

Graduate Research Assistant — Modeling Uni-Cellular Organism Movement Using a Hidden Markov Chain
California State University, Fullerton | PI: Dr. Nicholas Brubaker | Jan 2025– Aug 2025

- Researched long term non-uniform organism movement patterns by developing a stochastic process framework in MATLAB.
- Engineered a scalable Hidden Markov Model by introducing boundary conditions and generalizing the transition matrix.
- Analyzed the asymptotic behavior of the Markov chain using the eigenvalue analysis.
- Discovered that the organism tends to drift toward the boundary states over time.
- Successfully presented findings via an academic poster at a research symposium.

Undergraduate Research Assistant — Bounds on Arithmetic Rainbow Ramsey Multiplicities
California State University, San Bernardino | PI: Dr. Gabriel Elvin | June 2023 - May 2024

- Researched optimal labeling for an infinite one-dimensional array to maximize objective function solutions under constraints.
- Developed and executed Python algorithms to count solutions and optimize labeling across large arrays to identify convergence.
- Proved a key lemma used in a faculty research paper on convergence labeling and arithmetic bounds.
- Successfully communicated findings via an academics poster across 4 undergraduate research symposiums.

Math Tutor
California State University, San Bernardino, Tutoring Center | Nov 2022 - May 2024

- Tutored undergraduate students in mathematics coursework, including Calculus I/II, linear Algebra, Statistics, and Proofs.
- Integrated digital tools like Desmos and Geogebra to translate abstract mathematical concepts into visual and concrete examples.
- Commended in student reviews for creating a healthy learning environment that successfully reduced math anxiety and confidence.
- Managed an in-person and online portfolio of 50+ university students over a 2 year period, with a 70% retention rate.

SELECTED PROJECTS

Low-SWaP-Cost Passive Slotted Waveguide Array Antenna Design & Simulation
Master's Thesis | Industry-Sponsored Project | Aerospace Corporation | Jan 2026 - May 2026

- Spearheaded a 12 GHz passive slotted waveguide array antenna research simulation project for satellite/SAR applications.
- Diagnosed critical simulation bottlenecks utilizing the MATLAB Antenna Toolbox.
- Identified computing power limitation and excitation model constraints during waveguide antenna research.
- Engineered a custom workaround leveraging multi-core parallel processing and optimizing an integrated coaxial probe feed.
- Successfully presented findings to lead project manager, chief PhD engineer, and HR manager.

Satellite Constellation Solar Power Modeling Under Earth Occultation
Mathematical Modeling Course Project | Industry-Sponsored Project | Aerospace Corporation | Aug 2025 - Aug 2025

- Researched Earth Shadowing effects on satellite solar power generation.
- Utilizing a custom multi-body orbital mechanics simulator with tensor modeled satellite constellation variables in MATLAB.
- Modeled solar irradiance, solar panel power, and Earth's shadow (umbra and penumbra).
- Implemented a vector geometric logic algorithm to dynamically track satellite entry and exit within Earth's shadow.
- Simulated a low-orbit, 3-plane, 9-satellite constellation over a 10 day period to evaluate long term power generation behavior.
- Discovered that satellites that avoid Earth's shadow on average receive 13% more power and 90° panels gained 14% more power.

EDUCATION

California State University, Fullerton — Fullerton, CA
M.S. in Mathematics / Computational and Applied Mathematics, May 2026, GPA: **3.8/4.0**

California State University, San Bernardino — San Bernardino, CA
B.S. in Mathematics / Applied Mathematics, May 2024, GPA: **3.7/4.0**

Publication:
Bounds on Arithmetic Rainbow Ramsey Multiplicities, arXiv:2601.05442; under review at *PUMP Journal*, 2nd revisions pending